

Critical Analysis of Histologic Criteria for Grading Atypical (Dysplastic) Melanocytic Nevi

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Key Words: Melanocytic nevus; Histologic grading; Severe dysplasia; Regression; Atypical-mole syndrome

Abstract

Low concordance in grading atypical (dysplastic) melanocytic nevi (AMN) has been reported, and no systematic evaluation is available. We studied 123 AMN with architectural and cytologic atypia (40 associated with atypical-mole syndrome), classified according to standard criteria by 3 independent observers. Histologic variables included junctional and dermal symmetry, lateral extension, cohesion and migration of epidermal melanocytes, maturation, regression, nuclear features, nuclear grade, melanin, inflammatory infiltrate location, and fibroplasia. AMN (43 junctional and 80 compound) were graded mild (31), moderate (61), and severe (31). AMN-severe correlated with 3 or more nuclear abnormalities (especially pleomorphism, heterogeneous chromatin, and prominent nucleolus) and absence of regression, mixed junctional pattern, and suprabasilar melanocytes on top of lentiginous hyperplasia. AMN-severe diagnostic accuracy was 99.5% using these criteria, but only the absence of nuclear pleomorphism differentiated AMN-mild from AMN-moderate. No architectural features distinguishing AMN-mild from AMN-moderate were selected as significant by the discriminant analysis. AMN from atypical-mole syndrome revealed subtle architectural differences, but none were statistically significant in the discriminant analysis. Histologic criteria can reliably distinguish AMN-severe but fail to differentiate AMN-mild from AMN-moderate. AMN from atypical-mole syndrome cannot be diagnosed using pathologic criteria alone.

Atypical (dysplastic) melanocytic nevi (AMN) are an important marker for increased melanoma risk, demonstrating an 8-fold or greater association with malignant melanoma.^{1,2} One clinically atypical nevus has been associated with a 2-fold risk, while 10 or more confer a 12-fold increased risk.³ AMN histology is well characterized, allowing a reliable distinction from common acquired nevi and malignant melanoma,^{4,5} but the reported agreement on the degree of atypia is normally low.^{5,6} In addition, the cytologic atypia seen in such lesions has been suggested to be incidental to a proliferative radial growth rather than indicative of dysplasia.⁷

AMN have been defined mainly by architectural and cytologic criteria that include basilar proliferation of atypical melanocytes in a lentiginous or nested pattern and an association with the following: (1) the presence of lamellar fibrosis or concentric eosinophilic fibrosis, (2) neovascularization, (3) inflammatory response, and (4) fusion of rete ridges.⁸ These histopathologic characteristics have correlated objectively with nuclear atypia,^{1,9} improving the consensus and reproducibility of AMN diagnosis.¹⁰ Therefore, the minimal essential histologic criteria proposed for AMN must be based on both nuclear atypia and abnormal patterns of intraepidermal melanocytic proliferation.^{1,11}

Any grading system must be able to consistently recognize 2 or more grades of atypia with biologic relevance.¹² The subjective definition for grading of dysplasia in AMN has resulted in no consensus on that matter, resulting in a concordance ranging from 35% to 58% (kappa value, 0.38-0.47) for experienced dermatopathologists and from 16% to 65% (kappa value, 0.05-0.24) for those with less experience. However, a statistically significant relationship has been described between melanocytic dysplasia and the person's

total number of melanocytic lesions, indicating that the pathologic grading scheme may be useful.¹³ Further refinement of the criteria for grading melanocytic dysplasia and experience in grading are critical for accuracy in subcategorization of AMN.⁵ No systematic evaluation of the histologic criteria for grading AMN is available to date.

This article describes a statistically validated system for grading melanocytic dysplasia in a series of AMN, a third associated with atypical-mole syndrome, prioritizing the histologic features. We also performed a detailed comparison of histologic features in sporadic AMN with those associated with the atypical-mole syndrome.

Materials and Methods

We retrospectively studied 123 consecutive lesions clinically diagnosed as atypical melanocytic nevi from 111 patients biopsied between January 1, 1989, and December 31, 1998, in St Bartholomew's and the Royal London Hospitals, London, England. All nevi fulfilled the standard definition of AMN^{3,14}; the clinical records were reviewed for patient age, nevus location, and familial or personal history of atypical-mole syndrome or malignant melanoma. All specimens were fixed in 10% buffered formalin, routinely processed, serially sectioned, and stained with H&E. All nevi showed architectural and cytologic atypia and were evaluated histologically and graded by 3 independent observers (R.C., A.B., and S.J.D.-C.), according to standard criteria (see "Degree of atypia," below).^{1,5,8} In case of grading disagreement, the lesions were discussed during simultaneous inspection before final categorization. Reproducibility data were not recorded. No case had evidence of scarring, incomplete resection precluding adequate evaluation, or features suggesting congenital onset.

A standardized protocol was developed for reviewing each lesion, including the following histologic features:

1. Degree of atypia. The grading was evaluated on multiple H&E-stained routinely processed sections of the whole lesion. Standard criteria were used for that purpose.^{5,11} Those criteria included the presence of lentiginous melanocytic hyperplasia (from discontinuous to confluent), upward melanocyte migration (little or no to fully pagetoid spread), nesting variation and bridging (few nests to confluent nests), and nuclear features comprising enlargement (compared with basal keratinocyte nuclei), pleomorphism, presence of nucleoli, and hyperchromatism. All these variables were evaluated independently and considered positive if present in at least 50% of the sample. All AMN were required to show architectural and cytologic atypia (at least 1 criterion from each group) and were scored according to the number of criteria fulfilled as mild (2-3 criteria), moderate

(4-5 criteria), or severe (6-7 criteria). Processing artifacts were found to influence the cohesion of cells in nests (from preserved to diminished); therefore, this feature was excluded from the final classification.

2. Nevus type was registered according to standard criteria as junctional or compound.

3. Lateral extension, studied by the presence and pattern of melanocyte proliferation extending at least 3 rete ridges or "pegs" beyond any dermal component. The pattern at this level was classified with the same criteria used for the general junctional pattern.

4. General junctional pattern. The pattern was classified according to the predominant pattern as lentiginous (basilar melanocytic hyperplasia mainly), nested (melanocyte nest predominating), or mixed lentiginous-nested.

5. Junctional symmetry was judged by 2 separate issues: (1) symmetry evaluated by the number of involved rete ridges beyond the dermal ridge at both ends (rete ridge differences >3 were considered evidence of asymmetry) and (2) symmetry judged by the junctional pattern at both ends of the nevus (lentiginous, nested, and mixed). The presence of the same pattern at both sides was considered evidence of "side-to-side symmetry." Otherwise the lesion was registered as asymmetric.

6. Junctional nest features comprised orientation evaluated by the predominant pattern (>50% of the lesion) and cohesion. Only sharply demarcated melanocyte nests were considered cohesive, regardless of the cellular cohesion within the nests.

7. Presence of suprabasal melanocytes migrating upward beyond the epidermal basal layer in at least 2 high-power fields. The location of suprabasal melanocytes in relation to intraepidermal theques and lentiginous hyperplasia also was recorded.

8. Melanocyte maturation and nest size in the dermal component. A progressively decreased melanocyte and dermal nest size was considered evidence of maturation. Cell type maturation (round or spindle cell) also was considered.

9. Predominant cell type for junctional and dermal components (if applicable) was evaluated as described in No. 8.

10. Nuclear features. The following variables were analyzed independently and registered as positive if present in more than 50% of melanocytes:

- Size. The absolute assessment of nuclear size is reliable only by morphometric evaluation. Therefore, only anisokaryosis, which needs no external control, was studied for a more accurate analysis. Nuclear size variation more than 3:1 was considered evidence of anisokaryosis.

- Pleomorphism was defined as irregularity in nuclear contour associated with nuclear grooves. The presence of nuclear pseudoinclusions also was recorded.

- Chromatin distribution in fine and homogeneous (euchromatic) or gross and heterogeneous (hyperchromatism) patterns was registered.

- Presence of prominent nucleolus.

11. Nuclear grade was coded as high if at least 3 of the aforementioned nuclear features were suggestive of malignancy.

12. The presence of melanin granules in the cytoplasm of melanocytes was recorded along with granule size in finely divided granules (dusty cytoplasm) or large granules at least one third the size of an erythrocyte.

13. Location of inflammatory infiltrate. Lymphocytic infiltrate was coded by its predominant location in epidermal, perivascular, or interstitial locations. Lesions with no preferential location were registered as nonspecific.

14. Both presence/absence and degree of fibroplasia were used to evaluate the stromal changes. Fibroplasia was registered as concentric eosinophilic fibrous tissue or lamellar (delicately layered fibrous tissue with entrapped spindle-shaped cells).

15. Regression was identified histologically by melanocyte dropout, dermal fibrosis different from lamellar fibroplasia, increased vascularity in the dermis, and a variable degree of lymphocytic infiltrate with melanophages.

All variables were analyzed statistically using the Fisher exact test for qualitative variables and analysis of variance and the Student *t* test for quantitative variables. The data were compared for the degree of atypia, the specific distinction between AMN-mild and AMN-moderate, and the differentiation of AMN associated with atypical-mole syndrome. Differences were considered statistically significant if *P* < .05, and significant variables were used for a stepwise multivariate analysis.¹⁵

Results

AMN were found in 65 women and 46 men, mean $\pm$ SD age 36.56 $\pm$ 11.94 years, and they were located in arms (35 lesions), legs (31 lesions), abdomen (28 lesions), and back (21 lesions). Three lesions were found on the face; in 5 cases the location was unknown. All nevi met the criteria proposed for AMN^{1,8} and comprised 43 junctional and 80 compound melanocytic nevi, graded AMN-mild in 31 (25.2%), AMN-moderate in 61 (49.6%), and AMN-severe in 31 (25.2%).

Lateral extension was present in 79 of 80 compound AMN with a lentiginous pattern in 29 AMN (23.6%), nested in 41 AMN (33.3%), and mixed in 9 AMN (7.3%). The dermal component was asymmetrically located (difference in the lateral extension >3 rete ridges) in 23 AMN (18.7%). The junctional component was lentiginous in 50 AMN (40.6%), nested in 54 AMN (43.9%), and mixed in 19 AMN (15.4%),

all associated with junctional asymmetry) **Image 1A**, with horizontal theque orientation in 100 AMN (81.3%), vertical in 11 (8.9%), and mixed in 12 (9.7%). Junctional nests were considered cohesive in 87 AMN (70.7%) and noncohesive in 31 (25.2%), while in the remaining 5 AMN (4.1%), the absence of well-formed melanocyte theques precluded the evaluation of this feature. Suprabasal melanocytes were observed in 45 AMN (36.6%), on top of melanocyte theques only in 13 AMN (10.6%) and on top of lentiginous melanocytic hyperplasia in 32 AMN (26.0%) **Image 1B**.

The dermal component of AMN revealed melanocyte maturation in the deeper part of the nevus in 73 (91%) of 80 compound AMN (as small round cells in all cases), while maturation was lacking in 7 (9%) of 80 compound AMN. Dermal melanocyte nests were smaller at the deeper nevus compartment in 43 (54%) of 80 compound AMN and with the same average size in the superficial and deep compartments in 7 (9%) of 80 compound AMN.

The cytologic features revealed epithelioid cell predominance in 117 AMN (95.1%) and spindle cell predominance in 6 (4.9%). At least 1 nuclear abnormality was observed in each case: prominent pleomorphism in 54 AMN (43.9%), associated with pseudoinclusions in 22 AMN [17.8%], prominent anisokaryosis in 104 AMN (84.6%), hyperchromatism in 36 AMN (29.3%), and a prominent nucleolus in 67 AMN (54.5%). High nuclear grade was diffuse in all 31 AMN-severe lesions (25.2%) **Image 1C** and in occasional cells in 26 mildly or moderately atypical nevi. In the latter lesions, the serial sectioning consistently demonstrated high-grade changes in fewer than 5% of atypical melanocytes, not requiring any reclassification of the lesions. The cytoplasm displayed large melanin granules in 85 AMN (69.1%) and minimal to moderate amounts of finely granular melanin in 38 AMN (30.9%).

The stromal reaction included inflammation and fibroplasia. A variable lymphocytic infiltrate was identified in all lesions (with histiocytes if regression was associated), showing perivascular distribution in 73 AMN (59.3%), interstitial in 30 AMN (24.4%), and no preferential location in 16 AMN (13.0%). Four additional lesions (3.2%) with a nonspecific location of inflammatory infiltrate also revealed epidermal lymphocytic exocytosis. Fibroplasia was found in 117 AMN (95.1%), revealing a concentric distribution in 51 lesions (41.5%) and a lamellar distribution in 66 (53.6%). Histologic regression was identified in 85 AMN (69.1%) **Image 2**.

Statistical Analysis

Most of the histologic features analyzed allowed the distinction of AMN-severe from AMN-mild and AMN-moderate, while no positive finding reliably distinguished AMN-mild from AMN-moderate. Mainly architectural and

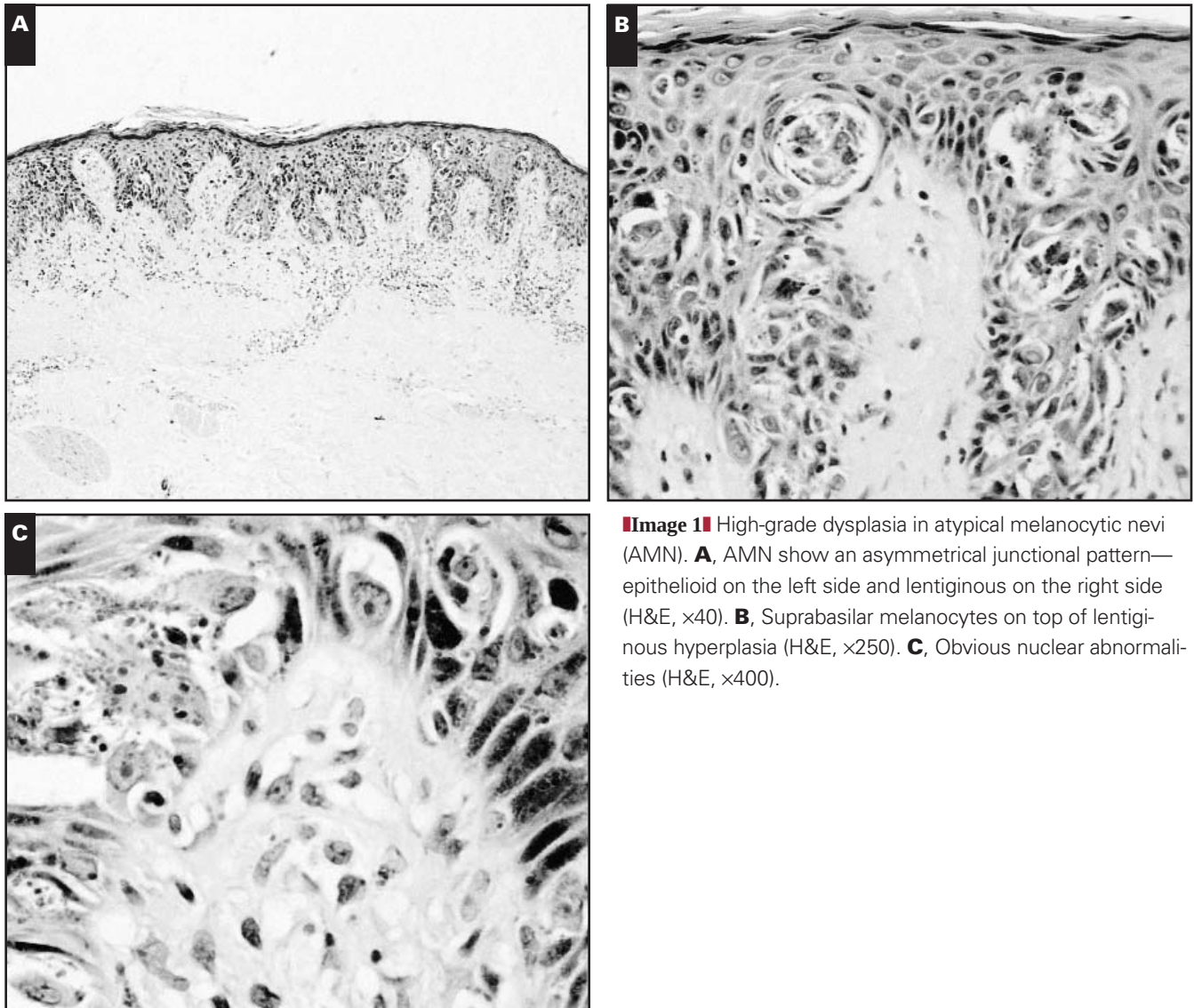


Image 1 High-grade dysplasia in atypical melanocytic nevi (AMN). **A**, AMN show an asymmetrical junctional pattern—epithelioid on the left side and lentiginous on the right side (H&E, ×40). **B**, Suprabasilar melanocytes on top of lentiginous hyperplasia (H&E, ×250). **C**, Obvious nuclear abnormalities (H&E, ×400).

growth patterns differentiated the latter 2 AMN types. However, no architectural features distinguishing AMN-mild from AMN-moderate were selected as significant by the discriminant analysis. Only 2 nuclear features differentiated AMN-mild from AMN-moderate: AMN-mild showed no nuclear pleomorphism and a lower incidence of anisokaryosis **Table 1**.

Multivariate analysis of significant variables selected 2 architectural features (junctional symmetry, presence and location of suprabasilar melanocytes) and 3 nuclear variables (pleomorphism, chromatin pattern, and nucleolus prominence) as major grading prognosticators **Figure 1** **Image 3**. The presence of histologic regression was a useful but less important grading variable. The probability of finding the 5 architectural and nuclear features in AMN-mild or moderate (1 – specificity) and of not finding them in AMN-severe is given in **Table 2**. The application of the 5 architectural and

nuclear features mentioned in the present series is shown in **Figure 2**. Remarkably, prominent nuclear pleomorphism was demonstrated in all AMN-severe but in no AMN-mild. In addition, broad overlap in the distribution of nuclear features was observed in AMN-mild and AMN-moderate.

No major criterion used for grading was found significantly different in AMN associated with the atypical-mole syndrome **Table 3**. All variables were architectural, and 4 of them also were useful for the distinction between AMN-mild and AMN-moderate. No nuclear feature distinguished AMN associated with the atypical-mole syndrome.

Discussion

Histologic criteria reliably distinguished AMN-severe from AMN-mild and AMN-moderate, based on the study of

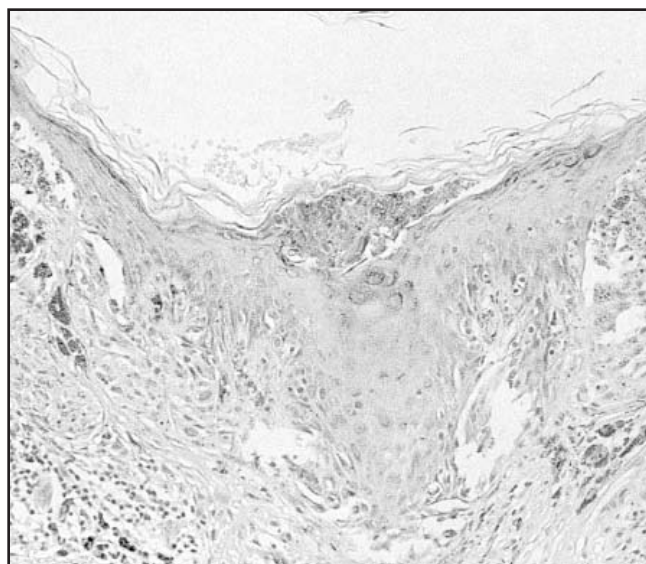


Image 2 Regression in low-grade atypical melanocytic nevi. This nevus shows numerous melanophages, prominent blood vessels, and lymphocytic infiltrate accounting for regression. Nuclear features corresponded to low-grade dysplasia (anisokaryosis and coarse chromatin; H&E, $\times 100$).

2 architectural and 3 major nuclear variables. Our systematic approach failed to differentiate between AMN-mild and AMN-moderate, two conditions that frequently share histologic features such as regression, suggesting a close association for AMN-mild and AMN-moderate. Therefore, we propose grouping AMN-mild and AMN-moderate as low-grade AMN in contrast with high-grade AMN (AMN-severe) (Figure 2). Only subtle architectural differences distinguished AMN associated with atypical-mole syndrome in the present series.

Both architectural and cytologic criteria were useful for the distinction of high-grade AMN in the present series and must be used together to define it. All lesions included fulfilled a widely accepted definition for AMN: basilar proliferation of atypical melanocytes organized in a lentiginous or nested pattern along with abnormal architectural patterns and stromal changes.^{1,8,11} Any grading system for melanocytic dysplasia must have biologic relevance and be able to consistently recognize 2 or more grades of atypia. The system currently used for grading AMN recognizes 2 groups: low-grade dysplasia (often graded as mild or slight) and high-grade dysplasia (often graded as moderate or severe),^{11,16} like the systems used for grading dysplasia in the uterine cervix¹⁷ and in inflammatory bowel disease.¹⁸ Many histologic features have been proposed for grading, but their definitions usually have been subjective and sometimes too vague, resulting in low concordance in AMN grading.^{5,6,19} In addition, cases reported are heterogeneous regarding the requirement of cytologic atypia.^{1,7,9,13}

The grading system we propose provides a clear definition with reliable histologic criteria useful for diagnosing high-grade AMN (AMN-severe), based on the study of 2 major architectural features (junctional symmetry, presence and location of suprabasal melanocytes) and 3 major nuclear variables (pleomorphism, chromatin pattern, and nucleolus prominence). The minimal requirements for the histologic diagnosis of high-grade AMN would include an atypical melanocyte proliferation with alterations in 1 architectural feature and 2 nuclear features (along with anisokaryosis) (Image 1 and Figure 2). This working definition stresses the importance of both architectural and cytologic atypia, prioritizing the analysis based on the most specific criteria (referred to as major in the present series). Anisokaryosis is a sort of general diagnostic criterion of AMN (present in 104/123 [84.6%] in this series),⁶ required to improve the specificity and avoid misclassification of a small proportion of low-grade AMN (1/61 [2%]; AMN-moderate in this series). The probability of finding major and anisokaryosis criteria in a low-grade AMN (Table 2, Figure 2) would be 0.054 (junctional pattern) $\times 0.250$ (nuclear pleomorphism) $\times 0.467$ (nucleolus) $\times 0.804$ (anisokaryosis) = 0.005 in the worst scenario, thus resulting in a diagnostic accuracy of 99.5% for high-grade AMN. This classification system would agree with the clinical system proposed by Kelly et al,¹⁴ who reported that a careful clinical examination of individual melanocytic nevi separates severe dysplasia from low-grade (mild or moderate) dysplasia in a high percentage of nevi from patients with the dysplastic nevus syndrome. A general application of clinical and histologic grading systems will improve the clinicopathologic correlation in AMN.

The grading system failed to distinguish AMN-mild from AMN-moderate. The features differentiating these two conditions were mainly architectural, some of them related to the association with the atypical-mole syndrome (see later text). A low incidence of nuclear abnormalities and the absence of prominent pleomorphism mainly characterized AMN-mild. However, a reliable classification system should include positive findings, lacking in this case for AMN-mild because anisokaryosis must be considered a general feature of AMN (it was present in 104/123 AMN [84.6%] in this series), as suggested in a previous study.⁶ A similar situation was found for the major architectural features in the distinction of AMN-mild and AMN-moderate: junctional symmetry revealed no differences, while the differences demonstrated in the presence of suprabasal melanocytes on top of lentiginous hyperplasia were only relative and not enough for reliable distinction. The application of these criteria in grading AMN is shown in the diagnostic algorithm of Figure 2.

Regression was associated significantly with low-grade AMN ($P < .001$), although no significant difference was

Table 1
Statistically Significant Features Useful for Grading Atypical (Dysplastic) Melanocytic Nevi (AMN)*

Variable	AMN			P
	Mild	Moderate	Severe	
Lateral extension pattern [†]				< .001
Lentiginous	4 (3.2)	19 (15.4)	6 (4.9)	
Nested	19 (15.4)	20 (16.3)	2 (1.6)	
Mixed	1 (0.8)	4 (3.2)	4 (3.2)	
Nonapplicable	7 (5.7)	18 (14.6)	19 (15.4)	
Junctional pattern ^{††}				< .001
Lentiginous	8 (6.5)	29 (23.6)	13 (10.6)	
Nested	22 (17.9)	28 (22.8)	4 (3.2)	
Mixed	1 (0.8)	4 (3.2)	14 (11.4)	
Junctional nest cohesion [†]				< .001
Yes	27 (22.0)	51 (41.5)	9 (7.3)	
No	1 (0.8)	9 (7.3)	21 (17.1)	
Nonapplicable	3 (2.4)	1 (0.8)	1 (0.8)	
Suprabasal melanocytes ^{††}				< .001
Nest only	2 (1.6)	9 (7.3)	2 (1.6)	
Nest and lentiginous hyperplasia	1 (0.8)	5 (4.1)	26 (21.1)	
Nonapplicable	28 (22.8)	47 (38.2)	3 (2.4)	
Nuclear pleomorphism ^{††}				< .001
Slight	31 (25.2)	38 (30.9)	0 (0.0)	
Prominent	0 (0.0)	23 (18.7)	31 (25.2)	
Anisokaryosis [†]				.001
Slight	12 (9.8)	6 (4.9)	1 (0.8)	
Prominent	19 (15.4)	55 (44.7)	30 (24.4)	
Chromatin pattern [†]				.001
Thin	26 (21.1)	50 (40.6)	11 (8.9)	
Granular	5 (4.1)	11 (8.9)	20 (16.3)	
Nucleolus [†]				.001
Inconspicuous	21 (17.1)	28 (22.8)	7 (5.7)	
Prominent	10 (8.1)	33 (26.8)	24 (19.5)	
Lymphocyte infiltrate location [†]				< .001
Nonspecific	4 (3.2)	3 (2.4)	9 (7.3)	
Epidermal	3 (2.4)	1 (0.8)	0 (0.0)	
Perivascular	21 (17.1)	42 (34.1)	10 (8.1)	
Interstitial	3 (2.4)	15 (12.2)	12 (9.8)	
Histologic regression				< .001
Yes	27 (22.0)	52 (42.3)	6 (4.9)	
No	4 (3.2)	9 (7.3)	25 (20.3)	
Anatomic site [†]				< .001
Arms	13 (11.3)	3 (2.6)	19 (16.5)	
Legs	6 (5.2)	19 (16.5)	6 (5.2)	
Abdomen	6 (5.2)	19 (16.5)	3 (2.6)	
Back	15 (13.0)	6 (5.2)	0 (0.0)	
Atypical-mole syndrome [†]				.006
Yes	22 (17.9)	9 (7.3)	9 (7.3)	
No	22 (17.9)	43 (35.0)	18 (14.6)	

* This table includes variables selected by discriminant analysis, features useful to distinguish AMN-mild from AMN-moderate, and regression. Data are given as number (percentage) based on the total of 123, except for anatomic site, for which the total was 115. Bold-faced numbers represent actual frequencies that were significantly different from the expected range.

[†] Features that distinguished AMN-mild from AMN-moderate.

^{††} Major criterion.

found between AMN-mild and AMN-moderate. Barnhill et al¹ highlighted the presence of melanophages in the inflammatory infiltrate of such lesions with a strong inverse correlation with nuclear abnormalities and a similar incidence to the regression in our series (65.7% vs 69.1%, respectively). However, to date, regression has never been correlated with the degree of atypia in AMN. The same inverse correlation between regression and both nuclear grade and degree of atypia was confirmed statistically for 2 nuclear variables only: pleomorphism and chromatin pattern. Considering the

high incidence of the other nuclear features and their frequent coexistence, a reliable diagnosis of high-grade AMN in the presence of regression must require marked pleomorphism and heterogeneous chromatin, along with at least 1 more nuclear and 1 more architectural abnormality. Different alternative explanations can be proposed for the association between inflammation, regression, and degree of dysplasia in AMN:

1. The inflammatory response is reactive to neoplastic changes in melanocytes, a kind of immunologic response to

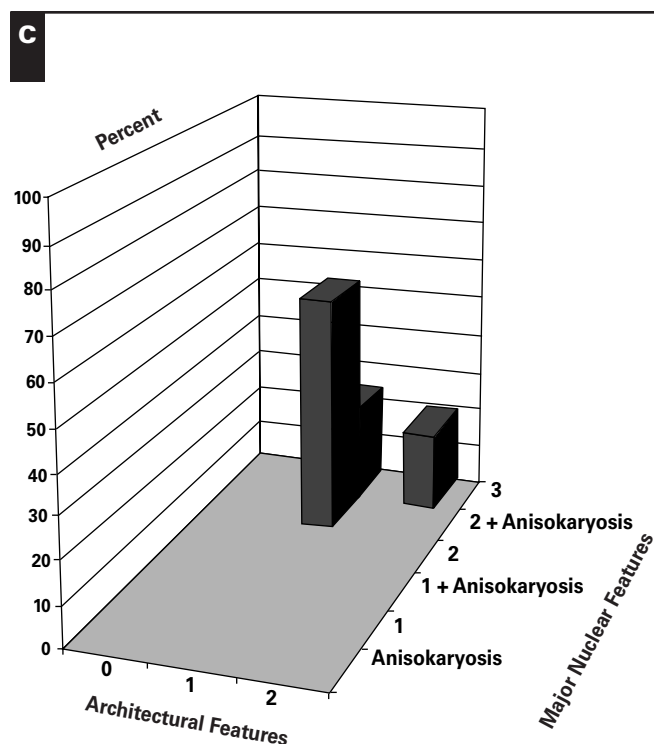
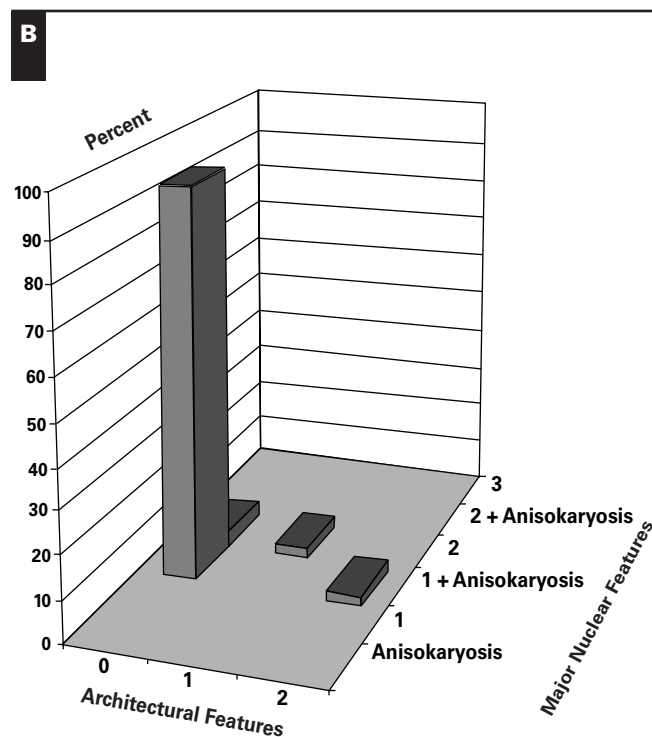
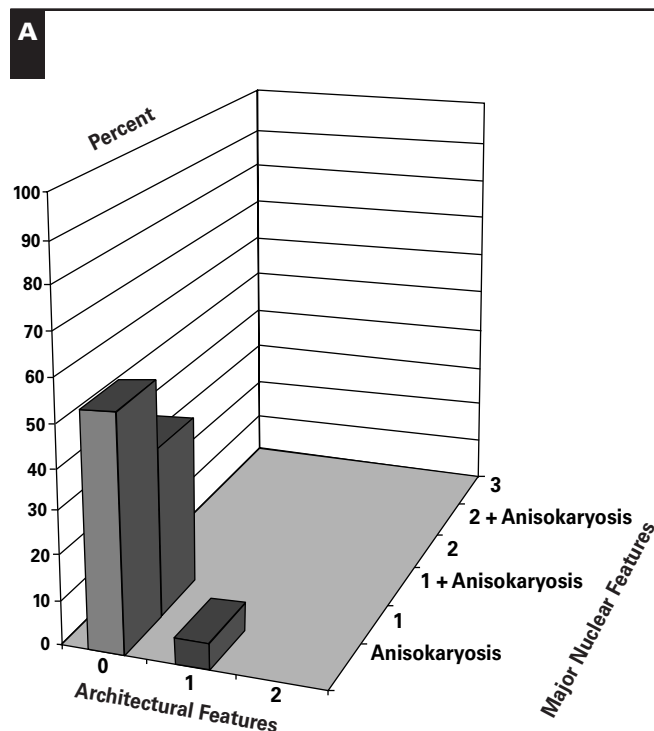


Figure 1 Distribution of major nuclear and architectural features by grading in atypical melanocytic nevi (AMN). **A**, AMN-mild showed 1 major nuclear abnormality or anisokaryosis; **B**, AMN-moderate, 2 major nuclear abnormalities or 1 major nuclear and anisokaryosis; and **C**, AMN-severe, 3 major nuclear abnormalities or 2 major nuclear abnormalities and anisokaryosis. Each bar represents the percentage of AMN with that feature.

the neoplasm. In that setting, we should expect a more intense inflammatory response in cases of severe nuclear atypia, in which the neoplastic transformation is fully established. Our results do not support that possibility.

2. The inflammatory response induces apoptosis, regression, and regenerative changes that result in nuclear atypia. Atypical nuclear changes due to apoptosis in other conditions,

such as adrenal cortical proliferation or pheochromocytoma, have been reported.^{20,21} These changes would be followed by an active phase of radial growth, resulting in the lentiginous hyperplasia and lateral expansion seen in AMN.⁷ The random atypia that is seen in AMN is suggested to be incidental to regressive and proliferative processes. In this scenario, the architectural atypia should be interpreted as the expression of

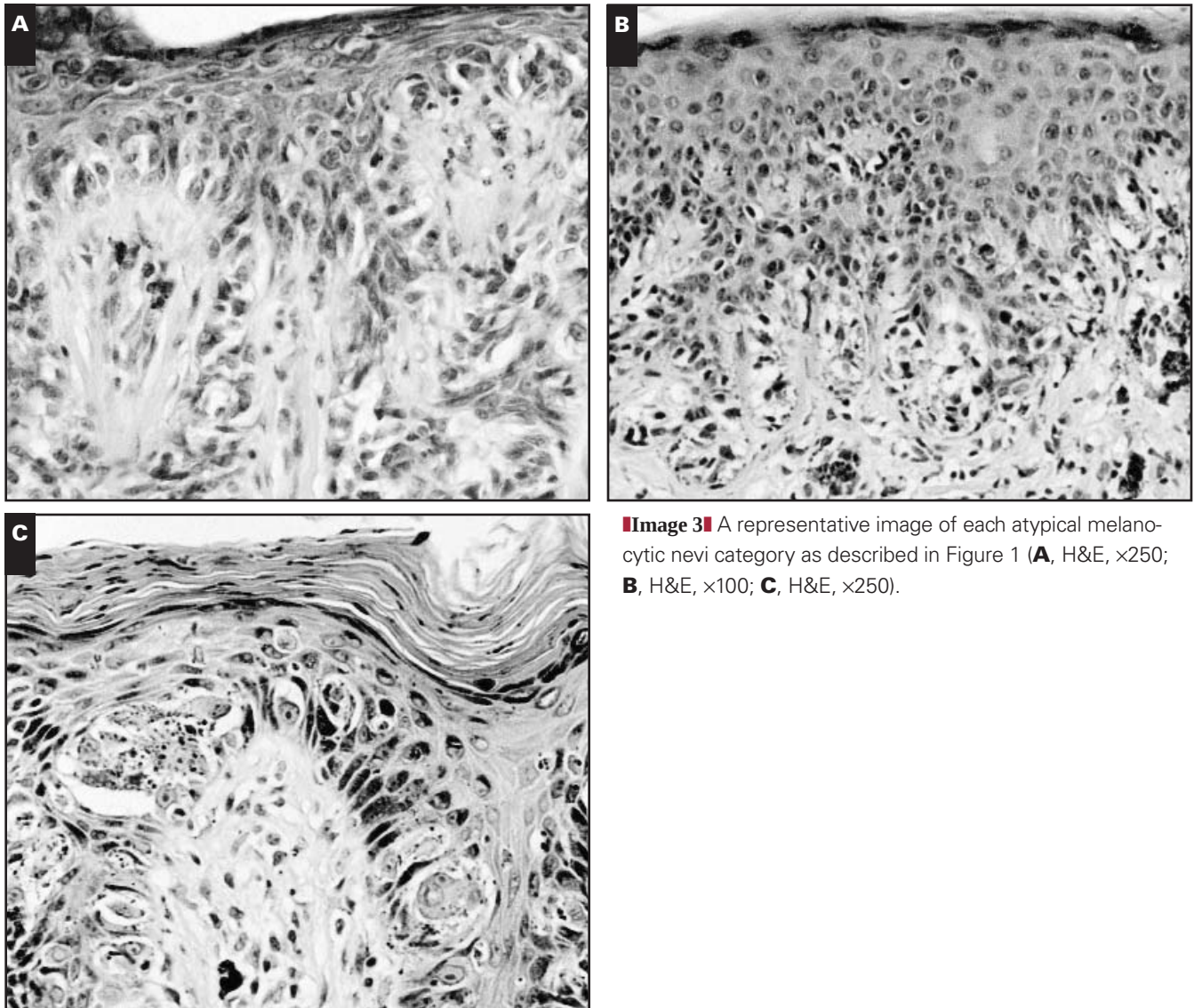


Image 3 A representative image of each atypical melanocytic nevi category as described in Figure 1 (**A**, H&E, $\times 250$; **B**, H&E, $\times 100$; **C**, H&E, $\times 250$).

tissue repair and would comprise melanocyte nest distortion and reactive fibroplasia, two common findings in AMN.

3. A common initiator is responsible for both melanocyte atypia and the inflammatory response.

Additional studies, including cell kinetics and genetic analyses, are needed to test the latter 2 hypotheses and the relative timing between melanocyte dysplasia and regression. Our results clearly identified AMN-severe as a different group in terms of the low incidence of regression features, the presence of extensive epidermal migration of melanocytes, and the presence of prominent nuclear pleomorphism (frequently associated with pseudoinclusions) (Table 1). These features suggest a neoplastic nature for this condition, while this question remains unanswered for AMN-mild and AMN-moderate.

Our results demonstrated a significant association between histologic regression and low-grade melanocytic

dysplasia. In low-grade AMN with regression, the active radial growth phase should be assumed reactive to the histologic regression. On the other hand, AMN-mild and AMN-moderate are heterogeneous lesions, and no histologic variable can reliably predict their potential evolution, even considering that long-term inflammatory responses may induce microsatellite instability, neoplastic transformation, and progression.²²

Subtle differences distinguished AMN associated with atypical-mole syndrome in the present series. To the best of our knowledge, our series represents the first systematic evaluation of histologic features in those AMN compared with other AMN not associated with the clinical syndrome. Familial melanoma has been associated with “clinically atypical moles,” which are markers for increased melanoma risk.^{2,3} The description of the atypical-mole syndrome marks the beginning of a continuing controversy on the real nature

Table 2
Accuracy in the Diagnosis of High-Grade Atypical (Dysplastic) Melanocytic Nevi (AMN) (AMN-Severe) by Significant Histologic Features*

Feature	Low-Grade AMN	High-Grade AMN
Junctional pattern		
Mixed	5	14
Nonmixed	87	17
Probability	5/92 = 0.054	17/31 = 0.548
Suprabasilar melanocytes		
On top of lentiginous hyperplasia	6	26
Other	86	5
Probability	6/92 = 0.065	5/31 = 0.161
Nuclear pleomorphism		
Prominent	23	31
Slight	69	0
Probability	23/92 = 0.250	0/31 = 0.000
Chromatin pattern		
Granular	16	20
Thin	76	11
Probability	16/92 = 0.174	11/31 = 0.355
Nucleolus		
Prominent	43	24
Inconspicuous	49	7
Probability	43/92 = 0.467	7/31 = 0.226
Anisokaryosis		
Prominent	74	30
Slight	18	1
Probability	74/92 = 0.804	1/31 = 0.032
Regression		
No	13	25
Yes	79	6
Probability	13/92 = 0.141	6/31 = 0.194

* Bold-faced numbers represent the variable values used to calculate the corresponding probability.

of histologically proven AMN,^{23,24} although the association of atypical-mole syndrome with increased risk of malignant melanoma is clear.^{2,25}

Our results mainly point out architectural differences in low-grade AMN for lesions associated with the atypical-mole syndrome (Table 3). These AMN frequently were located in the arms and were characterized by less cohesive proliferation of epithelioid melanocytes with a lower degree of atypia than in AMN not associated with that syndrome. There was no difference in the incidence of high-grade AMN in patients with the clinical syndrome, a finding consistent with the nature of high-grade AMN as a risk marker. Most AMN in patients with atypical-mole syndrome were low-grade (especially AMN-mild), probably related to the closer follow-up in these patients than in those without the syndrome. Although the differences were subtle in terms of frequency rather than qualitative, the architectural pattern seen in atypical-mole syndrome AMN suggests a cell adhesion disturbance in these nevi. That hypothesis would be consistent with the neoplastic progression pathway proposed for the sequence melanocytic nevus→melanocytic dysplasia→malignant melanoma.²⁶⁻²⁸ Down-regulation of E-cadherin may endow melanocytic cells with new adhesive

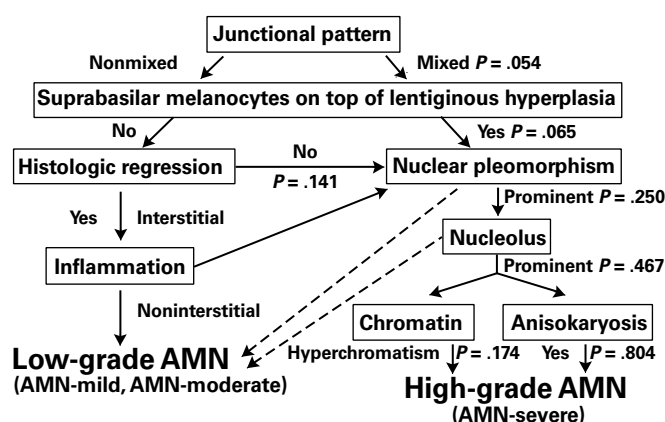


Figure 2 Diagnostic algorithm for grading atypical melanocytic nevi (AMN). Numbers represent the probability of finding that feature in low-grade AMN. The diagnostic accuracy dramatically improves if several histologic criteria are fulfilled.

properties and altered spatial relations that favor uncontrolled proliferation and migration in low-grade AMN,^{27,28} resulting in those subtle architectural changes.

High-grade AMN (AMN-severe) can be distinguished reliably from low-grade AMN (AMN-mild and AMN-moderate) based on the study of 2 architectural and 3 major nuclear variables. AMN-mild and AMN-moderate frequently share histologic features such as regression, suggesting a close association for AMN-mild and AMN-moderate and distinguishing them from high-grade AMN. Therefore, we propose grouping AMN-mild and AMN-moderate as low-grade AMN in contrast with high-grade AMN (AMN-severe). Finally, in our series, only subtle architectural differences distinguished AMN associated with atypical-mole syndrome, highlighting that the latter is a clinicopathologic diagnosis.

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Table 3

Statistically Significant Histologic Features for the Distinction of Atypical (Dysplastic) Melanocytic Nevi (AMN) From Atypical-mole Syndrome (AMS)*

Histologic Feature	AMS History		P
	Positive	Negative	
Degree of atypia			.006
AMN-mild	22 (17.9)	22 (17.9)	
AMN-moderate	9 (7.3)	43 (35.0)	
AMN-severe	9 (7.3)	18 (14.6)	
Lateral extension pattern			.022
Lentiginous	8 (6.5)	21 (17.0)	
Nested	16 (13.0)	25 (20.3)	
Mixed	0 (0.0)	9 (7.3)	
Nonapplicable	16 (13.0)	28 (22.8)	
Junctional nest orientation			.001
Horizontal	27 (22.0)	73 (59.3)	
Vertical	2 (1.6)	9 (7.3)	
Nonapplicable	11 (8.9)	1 (0.8)	
Junctional nest cohesion			.001
Yes	23 (18.6)	64 (52.0)	
No	13 (10.6)	18 (14.6)	
Nonapplicable	4 (3.2)	1 (0.8)	
Melanocyte maturation			.004
Yes	31 (25.2)	42 (34.1)	
No	0 (0.0)	7 (5.7)	
Nonapplicable	9 (7.3)	34 (27.6)	
Lymphocyte infiltrate location			.001
Nonspecific	9 (7.3)	7 (5.7)	
Epidermal	2 (1.6)	2 (1.6)	
Perivascular	26 (21.1)	47 (38.2)	
Interstitial	3 (2.4)	27 (22.0)	
Anatomic site			.002
Arms	19 (16.5)	16 (13.9)	
Legs	12 (10.4)	19 (16.5)	
Abdomen	3 (2.6)	25 (21.7)	
Back	6 (5.2)	15 (13.0)	

* Data are given as number (percentage) based on the total of 123, except for anatomic site, for which the total was 115. Bold-faced numbers represent actual frequencies that were significantly different from the expected range.

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